DAILY PRACTICE CALENDAR

Grade 10 Math: Geometry • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- This calendar follows the **California Common Core standards for Geometry**, the usual Grade 10 course. Students who want an extra challenge can add **Geometry Competition Math**. Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Points, Lines & Measurement
2	Aug 23–Aug 27	Angles & Logical Reasoning
3	Aug 30–Sep 3	Rigid Transformations
4	Sep 6–Sep 10	Symmetry & Compositions
5	Sep 13–Sep 17	Parallel Lines & Transversals
6	Sep 20–Sep 24	Proofs About Lines & Angles
7	Sep 27–Oct 1	Triangle Congruence
8	Oct 4–Oct 8	CPCTC & Isosceles Triangles
9	Oct 11–Oct 15	Bisectors, Medians & Midsegments
10	Oct 18–Oct 22	Review & Checkpoint 1
11	Oct 25–Oct 29	Parallelograms
12	Nov 1–Nov 5	Special Quadrilaterals & Coordinate Proofs
13	Nov 8–Nov 12	Dilations & Similarity
14	Nov 15–Nov 19	Similar Triangles
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Similarity in Right Triangles
16	Dec 6–Dec 10	Pythagorean Theorem & Special Right Triangles
17	Dec 13–Dec 17	Trigonometric Ratios
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Solving Right Triangles
19	Jan 10–Jan 14	Laws of Sines & Cosines
20	Jan 17–Jan 21	Mid-Year Review & Checkpoint
21	Jan 24–Jan 28	Arcs & Chords
22	Jan 31–Feb 4	Inscribed Angles
23	Feb 7–Feb 11	Tangents, Secants & Segment Lengths
24	Feb 14–Feb 18	Arc Length, Radians & Sectors
25	Feb 21–Feb 25	Equations of Circles
26	Feb 28–Mar 3	Volume & Cavalieri's Principle
27	Mar 6–Mar 10	Modeling with Geometry
28	Mar 13–Mar 17	Probability & Sample Spaces
29	Mar 20–Mar 24	Conditional Probability
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Independence & the Addition Rule
31	Apr 10–Apr 14	Counting & Probability
32	Apr 17–Apr 21	Year-End Review: Congruence, Similarity & Trig
33	Apr 24–Apr 28	Year-End Review & Practice Test
34	May 1–May 5	Final Checkpoint & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027 — Unit 1: Foundations & Transformations

Points, Lines & Measurement

Core: The distance and midpoint formulas measure segments on the coordinate plane.

Mon Aug 16

Segment addition

TRY B is between A and C. $AB = 2x + 1$, $BC = 3x - 4$ and $AC = 22$. Find x.

Tue Aug 17

Midpoint formula

TRY Find the midpoint of $(-4, 7)$ and $(6, -1)$.

Wed Aug 18

Distance formula

TRY Find the distance between $(-2, 3)$ and $(4, 11)$.

Thu Aug 19

Finding an endpoint

TRY $M(3, 2)$ is the midpoint of $\overline{AB}$, and A is $(-1, 5)$. Find B.

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027 — Unit 1: Foundations & Transformations

Angles & Logical Reasoning

Core: Name angle pairs correctly, and read if–then statements carefully.

Mon Aug 23

Angle addition

TRY $m\angle ABD = 70°$, $m\angle ABC = (3x + 4)°$ and $m\angle CBD = (2x + 6)°$. Find x.

Tue Aug 24

Linear pairs

TRY Two angles form a linear pair, and one is 4 times the other. Find both.

Wed Aug 25

Converses

TRY Write the converse of "If a figure is a square, then it has 4 sides." Is it true?

Thu Aug 26

Counterexamples

TRY Give a counterexample to "If $x^2 = 25$, then $x = 5$."

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027 — Unit 1: Foundations & Transformations

Rigid Transformations

Core: Translations, reflections and rotations keep every distance and angle the same.

Mon Aug 30

Translations

TRY Apply $(x, y) \to (x - 3, y + 5)$ to $(2, -1)$.

Tue Aug 31

Reflections

TRY Reflect $(4, -3)$ over the line $y = x$.

Wed Sep 1

Rotations

TRY Rotate $(5, 2)$ by $90°$ clockwise about the origin.

Thu Sep 2

Rotating about a point

TRY Rotate $(6, 1)$ by $180°$ about the point $(2, 3)$.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027 — Unit 1: Foundations & Transformations

Symmetry & Compositions

Core: Two transformations in a row can often be replaced by one.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Rotational symmetry

TRY What is the smallest rotation that maps a regular pentagon onto itself?

Tue Sep 7

Lines of symmetry

TRY How many lines of symmetry does a rectangle that is not a square have?

Wed Sep 8

Compositions

TRY Reflect $(3, 1)$ over the x-axis and then over the y-axis. What single transformation does the same thing?

Thu Sep 9

Two parallel mirrors

TRY Reflecting over two parallel lines 4 units apart is the same as what single transformation?

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Lines, Angles & Congruence

Parallel Lines & Transversals — **Core:** With parallel lines, corresponding and alternate angles are congruent.

Mon Sep 13
Alternate interior angles
TRY Alternate interior angles measure $(5x - 12)^\circ$ and $(3x + 20)^\circ$. Find x.

Tue Sep 14
Same-side interior angles
TRY Same-side interior angles measure $(2x + 10)^\circ$ and $(4x - 10)^\circ$. Find x.

Wed Sep 15
Proving lines parallel
TRY Two corresponding angles both measure 72°. What can you conclude?

Thu Sep 16
Parallel slopes
TRY Are $y = 3x - 2$ and $6x - 2y = 8$ parallel?

Fri Sep 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Lines, Angles & Congruence

Proofs About Lines & Angles — **Core:** Every step needs a reason: a definition, a postulate or a theorem.

Mon Sep 20
Vertical angles
TRY Why are vertical angles always congruent?

Tue Sep 21
Giving reasons
TRY Give the reason: if $\angle 1$ and $\angle 2$ form a linear pair, then $m\angle 1 + m\angle 2 = 180^\circ$.

Wed Sep 22
Perpendicular lines
TRY Two lines meet to form congruent adjacent angles. What does each measure?

Thu Sep 23
Triangle sum proof
TRY To prove a triangle's angles add to 180°, what line do you draw?

Fri Sep 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Lines, Angles & Congruence

Triangle Congruence — **Core:** SSS, SAS, ASA, AAS and HL prove congruence; SSA and AAA do not.

Mon Sep 27
Which shortcut?
TRY Two sides and the angle between them are congruent. Which shortcut applies?

Tue Sep 28
Not enough
TRY Do three pairs of congruent angles prove two triangles congruent?

Wed Sep 29
Right triangles
TRY Two right triangles have congruent hypotenuses and one pair of congruent legs. Are they congruent?

Thu Sep 30
Using congruence
TRY $\triangle ABC \cong \triangle DEF$, $AB = 2x + 3$ and $DE = 15$. Find x.

Fri Oct 1
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Lines, Angles & Congruence

CPCTC & Isosceles Triangles — **Core:** Corresponding parts of congruent triangles are congruent.

Mon Oct 4
CPCTC
TRY $\triangle PQR \cong \triangle XYZ$ and $m\angle Q = 48^\circ$. Find $m\angle Y$.

Tue Oct 5
Base angles
TRY An isosceles triangle has vertex angle 52°. Find each base angle.

Wed Oct 6
Isosceles algebra
TRY The base angles of an isosceles triangle are $(3x + 10)^\circ$ and $(5x - 20)^\circ$. Find x.

Thu Oct 7
Equilateral triangles
TRY Two sides of an equilateral triangle are $3y - 1$ and $2y + 4$. Find the side length.

Fri Oct 8
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Lines, Angles & Congruence

Bisectors, Medians & Midsegments **Core:** A midsegment is parallel to the third side and half as long.

Mon Oct 11
Midsegments
TRY A midsegment of a triangle is parallel to a side of length 18. How long is it?

Tue Oct 12
Perpendicular bisectors
TRY P is on the perpendicular bisector of $\overline{AB}$. $PA = 3x + 2$ and $PB = 5x - 6$. Find PA.

Wed Oct 13
The centroid
TRY A median of a triangle is 18 long. How far is the centroid from that vertex?

Thu Oct 14
The circumcenter
TRY Where is the circumcenter of a right triangle?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Lines, Angles & Congruence

Review & Checkpoint 1 **Core:** Pull Units 1–2 together and fix any weak spots.

Mon Oct 18
Review: transformations
TRY Reflect $(2, 7)$ over the x-axis, then translate it 3 units left.

Tue Oct 19
Review: parallel lines
TRY Corresponding angles measure $(4x + 8)^\circ$ and $(6x - 20)^\circ$. Find x.

Wed Oct 20
Review: congruence
TRY Is AAS a valid congruence shortcut?

Thu Oct 21
Review: midsegments
TRY A midsegment $2x + 1$ is parallel to a side $6x - 8$. Find x.

Fri Oct 22
Checkpoint 1
Units 1–2 test. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Quadrilaterals & Similarity

Parallelograms **Core:** Opposite sides and angles are congruent; the diagonals bisect each other.

Mon Oct 25
Angles
TRY One angle of a parallelogram is 65°. Find the other three.

Tue Oct 26
Diagonals
TRY The diagonals of $ABCD$ meet at E. $AE = 2x + 3$ and $EC = 5x - 9$. Find x.

Wed Oct 27
Proving a parallelogram
TRY If both pairs of opposite sides of a quadrilateral are congruent, must it be a parallelogram?

Thu Oct 28
Coordinates
TRY Parallelogram $ABCD$ has $A(0, 0)$, $B(5, 0)$ and $C(7, 3)$. Find D.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Quadrilaterals & Similarity

Special Quadrilaterals & Coordinate Proofs **Core:** Rectangles have congruent diagonals; rhombuses have perpendicular diagonals.

Mon Nov 1
Rectangles
TRY The diagonals of a rectangle are $3x + 1$ and $5x - 7$. Find x.

Tue Nov 2
Rhombuses
TRY A rhombus has diagonals 10 and 24. Find its side length.

Wed Nov 3
Trapezoids
TRY A trapezoid has bases 10 and 18. How long is its midsegment?

Thu Nov 4
Coordinate proof
TRY Is the quadrilateral with vertices $(0, 0)$, $(4, 1)$, $(5, 5)$, $(1, 4)$ a rhombus?

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Quadrilaterals & Similarity

Dilations & Similarity — **Core:** A dilation multiplies every distance from the center by the scale factor k.

Mon Nov 8
Dilations
TRY Dilate $(6, -4)$ by a scale factor of $\frac{1}{2}$ about the origin.

Tue Nov 9
Another center
TRY Dilate $(5, 3)$ by a scale factor of 2 about the point $(1, 1)$.

Wed Nov 10
Scale factor
TRY A side of length 6 has an image of length 15. Find the scale factor.

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Similar polygons
TRY Two similar polygons have sides in the ratio 3 : 5. The smaller perimeter is 24. Find the larger.

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Quadrilaterals & Similarity

Similar Triangles — **Core:** AA, SAS$\sim$ and SSS$\sim$ prove that two triangles are similar.

Mon Nov 15
AA similarity
TRY One triangle has angles $35°$ and $80°$; another has $80°$ and $65°$. Are they similar?

Tue Nov 16
SSS similarity
TRY Are triangles with sides 6, 8, 10 and 9, 12, 15 similar?

Wed Nov 17
Side-splitter theorem
TRY In $\triangle ABC$, $DE \parallel BC$. $AD = 6$, $DB = 4$ and $AE = 9$. Find EC.

Thu Nov 18
Indirect measurement
TRY A mirror is 2 m from a person whose eyes are 1.6 m high, and 15 m from a tree. The person sees the treetop in the mirror. How tall is the tree?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Quadrilaterals & Similarity

Similarity in Right Triangles — **Core:** The altitude to the hypotenuse splits a right triangle into two similar triangles.

Mon Nov 29
Geometric mean
TRY Find the geometric mean of 4 and 9.

Tue Nov 30
The altitude
TRY The altitude to the hypotenuse splits it into lengths 3 and 12. How long is the altitude?

Wed Dec 1
Finding a leg
TRY The hypotenuse is 25, and the part next to one leg is 9. How long is that leg?

Thu Dec 2
Area ratios
TRY Two similar triangles have scale factor 2 : 3. The smaller has area 20. Find the larger area.

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Right Triangles & Trigonometry

Pythagorean Theorem & Special Right Triangles — **Core:** 45-45-90 sides are $1 : 1 : \sqrt{2}$; 30-60-90 sides are $1 : \sqrt{3} : 2$.

Mon Dec 6
Pythagorean triples
TRY A right triangle has legs 20 and 21. Find the hypotenuse.

Tue Dec 7
45-45-90 triangles
TRY A 45-45-90 triangle has legs 7. Find the hypotenuse.

Wed Dec 8
30-60-90 triangles
TRY A 30-60-90 triangle has hypotenuse 12. Find both legs.

Thu Dec 9
Acute, right or obtuse?
TRY Is a triangle with sides 8, 15 and 16 acute, right or obtuse?

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Right Triangles & Trigonometry

Trigonometric Ratios

Core: SOH-CAH-TOA: sine, cosine and tangent are ratios of sides.

Mon Dec 13
Finding ratios
TRY A right triangle has legs 8 and 15 and hypotenuse 17. Find $\sin A$, where A is opposite the side 8.

Tue Dec 14
Cofunctions
TRY If $\sin 40^\circ \approx 0.643$, what is $\cos 50^\circ$?

Wed Dec 15
Exact values
TRY Find $\sin 60^\circ$ and $\tan 45^\circ$.

Thu Dec 16
Finding a side
TRY A right triangle has hypotenuse 10 and a 30° angle. Find the side opposite that angle.

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Right Triangles & Trigonometry

Solving Right Triangles

Core: Use inverse trigonometry to find angles.

Mon Jan 3
Finding an angle
TRY A right triangle has both legs equal to 5. Find the acute angles.

Tue Jan 4
Angle of elevation
TRY From 50 m away, the top of a building is at a 60° angle of elevation. How tall is the building?

Wed Jan 5
Angle of depression
TRY From the top of a 30 m lighthouse, a boat is at a 30° angle of depression. How far away is the boat?

Thu Jan 6
Ramps
TRY A ramp rises 1 ft for every 12 ft across. Find its angle to the nearest degree.

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Right Triangles & Trigonometry

Laws of Sines & Cosines

Core: Use these laws for triangles that are not right triangles.

Mon Jan 10
Law of sines
TRY In $\triangle ABC$, $A = 30^\circ$, $B = 45^\circ$ and $a = 10$. Find b.

Tue Jan 11
Law of cosines
TRY In $\triangle ABC$, $a = 7$, $b = 8$ and $C = 60^\circ$. Find c.

Wed Jan 12
Finding an angle
TRY A triangle has sides 5, 7 and 8. Find the angle opposite the side 7.

Thu Jan 13
Area with sine
TRY Find the area of a triangle with sides 8 and 10 and a 150° angle between them.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Right Triangles & Trigonometry

Mid-Year Review & Checkpoint

Core: Review Units 1–4 and practice working steadily through a full test.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: isosceles triangles
TRY An isosceles triangle has base angles of 70°. Find the vertex angle.

Tue Jan 18
Review: similarity
TRY In $\triangle ABC$, $DE \parallel BC$, $AD = 4$, $DB = 6$ and $AE = 6$. Find EC.

Wed Jan 19
Review: trigonometry
TRY In a right triangle, $\cos A = \frac{3}{5}$. Find $\tan A$.

Thu Jan 20
Review: special right triangles
TRY A square has diagonal 10. Find its side.

Fri Jan 21
Mid-Year Checkpoint
Units 1–4 test. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Circles

Arcs & Chords

Core: A central angle equals its arc; congruent chords cut off congruent arcs.

Mon Jan 24
Arc measures
TRY A central angle measures $110°$. Find the minor and major arcs.

Tue Jan 25
Chord distance
TRY A circle has radius 13. How far from the center is a chord of length 24?

Wed Jan 26
Equidistant chords
TRY Two chords are the same distance from the center. One has length 10. Find the other.

Thu Jan 27
Adjacent arcs
TRY Arcs $AB = (4x)°$ and $BC = (5x)°$ together form a semicircle. Find x.

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Circles

Inscribed Angles

Core: An inscribed angle is half of the arc it cuts off.

Mon Jan 31
Inscribed angles
TRY An inscribed angle cuts off an arc of $140°$. Find the angle.

Tue Feb 1
Angles in a semicircle
TRY What is the measure of an angle inscribed in a semicircle?

Wed Feb 2
Cyclic quadrilaterals
TRY In cyclic quadrilateral $ABCD$, $m\angle A = (2x + 10)°$ and $m\angle C = (3x - 5)°$. Find x.

Thu Feb 3
Tangent–chord angles
TRY A tangent and a chord meet on the circle and cut off an arc of $100°$. Find the angle.

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Circles

Tangents, Secants & Segment Lengths

Core: A tangent is perpendicular to the radius; two tangents from one point are congruent.

Mon Feb 7
Tangent length
TRY Point P is 17 from the center of a circle of radius 8. Find the tangent length from P.

Tue Feb 8
Equal tangents
TRY Tangents from P touch the circle at A and B. $PA = 3x - 4$ and $PB = x + 10$. Find PA.

Wed Feb 9
Crossing chords
TRY Chords AB and CD meet at P. $AP = 6$, $PB = 4$ and $CP = 3$. Find PD.

Thu Feb 10
Secant angles
TRY Two secants from an outside point cut off arcs of $120°$ and $40°$. Find the angle between them.

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Circles

Arc Length, Radians & Sectors

Core: Arcs and sectors are fractions of the circle; in radians, $s = r\theta$.

Mon Feb 14
Arc length
TRY Find the length of a $60°$ arc in a circle of radius 12.

Tue Feb 15
Sector area
TRY Find the area of a $120°$ sector of a circle with radius 6.

Wed Feb 16
Radians
TRY Convert $135°$ to radians.

Thu Feb 17
Using $s = r\theta$
TRY A circle has radius 5. How long is the arc for a central angle of 2 radians?

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Circles

Equations of Circles — **Core:** A circle with center (h, k) and radius r: $(x-h)^2 + (y-k)^2 = r^2$.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Writing equations

TRY Write the equation of the circle with center $(2, -3)$ and radius 5.

Tue Feb 22

Reading equations

TRY Find the center and radius of $(x+1)^2 + (y-4)^2 = 49$.

Wed Feb 23

Completing the square

TRY Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$.

Thu Feb 24

On the circle?

TRY Is $(5, 1)$ on the circle $(x-2)^2 + (y+3)^2 = 25$?

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 6: Volume, Modeling & Probability

Volume & Cavalieri's Principle — **Core:** Solids with equal heights and equal cross sections have equal volumes.

Mon Feb 28

Cylinders

TRY Find the volume of a cylinder with radius 4 and height 9.

Tue Feb 29

Cones

TRY Find the volume of a cone with radius 6 and height 10.

Wed Mar 1

Spheres

TRY Find the volume of a sphere with radius 6.

Thu Mar 2

Cavalieri's principle

TRY One stack of 20 coins is straight; another is slanted. Do they have the same volume?

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Volume, Modeling & Probability

Modeling with Geometry — **Core:** Model real objects with shapes; density = amount ÷ area or volume.

Mon Mar 6

Population density

TRY A town of 12,000 people covers 3 square miles. Find its population density.

Tue Mar 7

Mass from density

TRY A metal cube has edge 5 cm and density 8 g/cm^3. Find its mass.

Wed Mar 8

Designing a can

TRY A cylindrical can has radius 4 cm and must hold 320π cm^3. How tall must it be?

Thu Mar 9

Solids of rotation

TRY A 3×5 rectangle spins around its side of length 5. Name the solid and find its volume.

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Volume, Modeling & Probability

Probability & Sample Spaces — **Core:** Use sets: "or" is a union, "and" is an intersection, "not" is a complement.

Mon Mar 13

Sample spaces

TRY How many outcomes are there when 3 coins are flipped?

Tue Mar 14

Complements

TRY Three coins are flipped. Find P(at least one head).

Wed Mar 15

Unions

TRY A card is drawn from a standard 52-card deck. Find P(heart or king).

Thu Mar 16

Intersections

TRY For the same deck, find P(heart and king).

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Volume, Modeling & Probability

Conditional Probability **Core:** $P(B \mid A) = P(A \text{ and } B) \div P(A)$.

Mon Mar 20
From a table
TRY Of 100 students, 40 play a sport, and 25 of those also play music. Find P(music | sport).

Tue Mar 21
Using the formula
TRY If $P(A) = 0.5$ and $P(A \text{ and } B) = 0.2$, find $P(B \mid A)$.

Wed Mar 22
Without replacement
TRY A bag has 2 red and 3 blue marbles. Two are drawn. Find P(second is red | first is red).

Thu Mar 23
Reading the condition
TRY 45 of 60 seniors drive and 10 of 40 juniors drive. A driver is picked at random. Find P(senior | drives).

Fri Mar 24
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Volume, Modeling & Probability

Independence & the Addition Rule **Core:** Independent: $P(A \text{ and } B) = P(A) \times P(B)$. Either: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

Mon Apr 3
Independent events
TRY A and B are independent, $P(A) = 0.3$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

Tue Apr 4
Testing independence
TRY $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \text{ and } B) = 0.2$. Are A and B independent?

Wed Apr 5
The addition rule
TRY $P(A) = 0.6$, $P(B) = 0.3$ and $P(A \text{ and } B) = 0.1$. Find $P(A \text{ or } B)$.

Thu Apr 6
Mutually exclusive events
TRY A die is rolled. Find P(1 or an even number).

Fri Apr 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Volume, Modeling & Probability

Counting & Probability **Core:** Use permutations when order matters and combinations when it does not.

Mon Apr 10
Permutations
TRY How many ways can a president, vice president and secretary be chosen from 8 people?

Tue Apr 11
Combinations
TRY How many 3-person committees can be chosen from 8 people?

Wed Apr 12
Probability with combinations
TRY Two students are chosen from 4 boys and 3 girls. Find P(both are girls).

Thu Apr 13
Arrangements
TRY How many ways can the letters of GEOMETRY be arranged?

Fri Apr 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Year-End Review

Year-End Review: Congruence, Similarity & Trig **Core:** Mix the skills from Units 1–4 and show your reasoning.

Mon Apr 17
Review: congruence
TRY Two angles and a side not between them are congruent. Which shortcut applies?

Tue Apr 18
Review: right-triangle similarity
TRY The altitude to the hypotenuse splits it into lengths 4 and 16. How long is the altitude?

Wed Apr 19
Review: trigonometry
TRY From 20 m away, the top of a pole is at a $45°$ angle of elevation. How tall is the pole?

Thu Apr 20
Timed set: 10 in 20 min
TRY Find each interior angle of a regular hexagon.

Fri Apr 21
Weekly Quiz
10 mixed review problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Year-End Review

Year-End Review & Practice Test

Core: Practice a full test, then learn from every error.

Mon Apr 24

Review: circles

TRY Find the center and radius of $x^2 + y^2 + 8y = 9$.

Tue Apr 25

Review: volume

TRY Find the volume of a hemisphere with radius 3.

Wed Apr 26

Review: probability

TRY If $P(A) = 0.7$ and $P(B \mid A) = 0.4$, find $P(A \text{ and } B)$.

Thu Apr 27

Practice Test

Full-length Geometry practice test covering the whole year.

Fri Apr 28

Error-log review

Redo every missed problem from the test; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Year-End Review

Final Checkpoint & Celebration

Core: Show what you have learned and get ready for Algebra 2.

Mon May 1

Light review

TRY Find the midpoint of $(-2, 8)$ and $(6, 0)$.

Tue May 2

Mixed warm-up set

TRY An inscribed angle cuts off an arc of $84°$. Find the angle.

Wed May 3

Final Checkpoint

Covers the whole year.

Thu May 4

Checkpoint review

Go over solutions together; students explain one problem each.

Fri May 5

Celebration & reflection

Certificates, progress review, and goals for Algebra 2 next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: $x = 5$
Tue: $(1, 3)$
Wed: 10
Thu: $(7, -1)$

Week 2
Mon: $x = 12$
Tue: $36°$ and $144°$
Wed: "If a figure has 4 sides, it is a square." False
Thu: $x = -5$

Week 3
Mon: $(-1, 4)$
Tue: $(-3, 4)$
Wed: $(2, -5)$
Thu: $(-2, 5)$

Week 4
Mon: $72°$
Tue: 2
Wed: $(-3, -1)$; a $180°$ rotation about the origin
Thu: A translation of 8 units

Week 5
Mon: $x = 16$
Tue: $x = 30$
Wed: The lines are parallel
Thu: Yes (both have slope 3)

Week 6
Mon: Both are supplementary to the same angle
Tue: Linear Pair Postulate
Wed: $90°$
Thu: A line through one vertex parallel to the opposite side

Week 7
Mon: SAS
Tue: No (AAA gives similarity only)
Wed: Yes, by HL
Thu: $x = 6$

Week 8
Mon: $48°$
Tue: $64°$
Wed: $x = 15$
Thu: 14

Week 9
Mon: 9
Tue: 14
Wed: 12
Thu: At the midpoint of the hypotenuse

Week 10
Mon: $(-1, -7)$
Tue: $x = 14$
Wed: Yes
Thu: $x = 5$

Week 11
Mon: $115°$, $65°$, $115°$
Tue: $x = 4$
Wed: Yes
Thu: $(2, 3)$

Week 12
Mon: $x = 4$
Tue: 13
Wed: 14
Thu: Yes: all four sides are $\sqrt{17}$

Week 13
Mon: $(3, -2)$
Tue: $(9, 5)$
Wed: 2.5
Thu: 40

Week 14
Mon: Yes
Tue: Yes (scale factor 1.5)
Wed: 6
Thu: 12 m

Week 15
Mon: 6
Tue: 6
Wed: 15
Thu: 45

Week 16
Mon: 29
Tue: $7\sqrt{2}$
Wed: 6 and $6\sqrt{3}$
Thu: Acute $(64 + 225 > 256)$

Week 17
Mon: $\frac{8}{17}$
Tue: About 0.643
Wed: $\frac{\sqrt{3}}{2}$ and 1
Thu: 5

Week 18
Mon: Both $45°$
Tue: $50\sqrt{3} \approx 86.6$ m
Wed: $30\sqrt{3} \approx 52$ m
Thu: $5°$

Week 19
Mon: $10\sqrt{2} \approx 14.1$
Tue: $\sqrt{57} \approx 7.55$
Wed: $60°$
Thu: 20

Week 20
Mon: $40°$
Tue: 9
Wed: $\frac{4}{3}$
Thu: $5\sqrt{2}$

Week 21
Mon: $110°$ and $250°$
Tue: 5
Wed: 10
Thu: $x = 20$

Week 22
Mon: $70°$
Tue: $90°$
Wed: $x = 35$
Thu: $50°$

Week 23
Mon: 15
Tue: 17
Wed: 8
Thu: $40°$

Week 24
Mon: 4π
Tue: 12π
Wed: $\frac{3\pi}{4}$
Thu: 10

Week 25
Mon: $(x - 2)^2 + (y + 3)^2 = 25$
Tue: Center $(-1, 4)$, radius 7
Wed: Center $(2, -3)$, radius 5
Thu: Yes $(9 + 16 = 25)$

Week 26
Mon: 144π
Tue: 120π
Wed: 288π
Thu: Yes

Week 27
Mon: 4,000 people per mi^2
Tue: 1,000 g
Wed: 20 cm
Thu: A cylinder; 45π

Week 28
Mon: 8
Tue: $\frac{7}{8}$
Wed: $\frac{4}{13}$
Thu: $\frac{1}{52}$

Week 29
Mon: $\frac{5}{8}$
Tue: 0.4
Wed: $\frac{1}{4}$
Thu: $\frac{9}{11}$

Week 30
Mon: 0.15
Tue: Yes
Wed: 0.8
Thu: $\frac{2}{3}$

Week 31
Mon: 336
Tue: 56
Wed: $\frac{1}{7}$
Thu: 20,160

Week 32
Mon: AAS
Tue: 8
Wed: 20 m
Thu: $120°$

Week 33
Mon: Center $(0, -4)$, radius 5
Tue: 18π
Wed: 0.28

Week 34
Mon: $(2, 4)$
Tue: $42°$

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